Career Launcher MBA

QA Algebra-05

Modulus; Logarithms

Question 1

If $7 + 3|x| = 13$, find the value of x.

(1) -2.5

(2) ± 6

(3) ± 2

(4) $\pm \frac{7}{2}$

Question 2

If $|x^2 + 1| \leq 1$, then find the value of $(x + 2)(x + 1)(x)(x - 1)(x - 2)$ .... 1000 terms

(1) –2

(2) 0

(3) –1

(4) > 0

Question 3

If $|3x| + 7|x| \leq 60$. Find the value of x.

(1) $-7 < x < 7$

(2) $x < 6$

(3) $-6 < x < 6$

(4) $-6 \leq x \leq 6$

Question 4

If a and b are positive integers such that $|a - 2| \leq 3$ and $|b + 3| \leq 4$. What is the minimum value of a^2/b^2?

(1) 1/49

(2) 25/49

(3) 1

(4) 4/49

Question 5

If $3x + 2|y| + y = 7$ and $x + |x| + 3y = 1$, then $x + 2y$ is

(1) –4/3

(2) 1

(3) 0

(4) 8/3

Question 6

The largest real value of a for which the equation $|x + a| + |x - 1| = 2$ has an infinite number of solutions for x is

(1) 0

(2) 2

(3) –1

(4) 1

For Questions 7 and 8

Answer the questions on the basis of the information given below.

$|x| + |y| = 7$, $|x|^2 + |y|^2 = 25$ $(x, y \in R)$

1. How many values of $x^3 + y^3$ are possible?

   (1) 1 (2) 2

   (3) 3 (4) 4

2. What is the value of $|x - y|^2$?

   (1) 1

   (2) 49

   (3) 4

   (4) Cannot be determined

Question 9

Find the number of integral values of x that satisfy the inequality $||2x - 19| - 7| < 5$.

(1) 13

(2) 12

(3) 11

(4) 10

Question 10

The number of integers n that satisfy the inequalities $|n - 60| < |n - 100| < |n - 20|$ is

(1) 19

(2) 18

(3) 20

(4) 21

Question 11

For a real number x the condition $|3x - 20| + |3x - 40| = 20$ necessarily holds if

(1) $9 < x < 14$

(2) $6 < x < 11$

(3) $10 < x < 15$

(4) $7 < x < 12$

Question 12

What is the minimum value of the function?

(1) $F(x) = |x - 1| + |x - 2| + |x - 4|$

(2) $F(x) = |x - 1| + |x - 2| + |x - 4| + |x - 8|$

Question 13

The number of distinct pairs of integers (m, n) satisfying $|1 + mn| < |m + n| < 5$ is

Question 14

Let $0 \le a \le x \le 100$ and $f(x) = |x - a| + |x - 100| + |x - a - 50|$.

Then the maximum value of f(x) becomes 100 when a is equal to

(1) 0

(2) 50

(3) 100

(4) 25

Question 15

What is the area bounded by the graphs

(1) $|x| + |y| = 10$

(2) $|x + 2| + |y – 3| = 10$

(3) $|x – y| + |x + y| = 10$

Question 16

If a, b, c and d are positive numbers such that $a^2 = b^2 = c^5 = d^6$, then the value of $\log_d(abc)$ is

(1) 5.8

(2) 6.0

(3) 7.2

(4) 6.4

Question 17

The value of $\left(\frac{1}{\log_5 210}+\frac{1}{\log_6 210}+\frac{1}{\log_7 210}\right)$ is

Question 18

If $\log_2 \log_3 \log_2 \log_x 2^{1024} = 1$, find the value of x?

Question 19

Solve for x if $3\log_{\frac{(x+4)}{8}}(x-1) = \log_{\frac{(x+4)}{8}} 27$.

(1) 4

(2) 3

(3) 2

(4) No solution

Question 20

If $P = 27^{\log_3 48} \times 3^{\log_{27} 48^2} \times 27^{\log_3 48^3} \times 3^{\log_{27} 48^4} \times \ldots \times 27^{\log_3 48^{99}} = 48^x$,

then find the value of 3x/50.

(1) 497

(2) 499

(3) 501

(4) 503

Question 21

How many real values of ‘x’, where ‘x’ is a real number, satisfy the equation

$81^{\log_{10} x} = 3 + 2x^{\log_{10} 9}$?

(1) 0

(2) 1

(3) 2

(4) 5

Question 22

Find the number of different values of x that satisfy the equation

$$2\left[27^{\log_3(\log_4 x)}\right] = (\log_4 x)^2 + 5(\log_4 x) + 2.$$

(1) 1

(2) 2

(3) 3

(4) 4

Question 23

How many ‘0’s are there in 3^{-70} that appear between the decimal and the first non-zero digit after the decimal. (Log 3 = 0.477)

(1) 34

(2) 33

(3) 30

(4) 35

Question 24

If $x \geq y$ and $y > 1$, then the value of the expression $\log_x\left(\frac{x}{y}\right) + \log_y\left(\frac{y}{x}\right)$ can never be.

(1) – 1

(2) – 0.5

(3) 0

(4) 1

Question 25

If $\log_2 x + \log_x 2 = 10/3 = \log_2 y + \log_y 2$ and $x \neq y$, then $x + y$ is equal to

(1) 2

(2) 65/8

(3) 10/3

(4) None of the above

Question 26

If $\log_7 \log_5 \left(\sqrt{x+5}+\sqrt{x}\right)=0$, what is the value of x?

(1) 2

(2) 3

(3) 4

(4) 5

Question 27

If $\log_{10}x - \log_{10}\sqrt{x} = 2\log_x 10$, then the possible value of x is given by

(1) 10

(2) 1/100

(3) 1/1000

(4) None of these

Question 28

If logyx = (a.logzy) = (b.logxz) = ab, then which of the following pairs of values for (a, b) is not possible?

(1) (–2, 1/2)

(2) (1, 1)

(3) (0.4, 2.5)

(4) (2, 2)

Question 29

If $\log_2 (5 \times 2^x + 1)$, $\log_4 (2^{1-x} + 1)$ and 1 are in A.P., then x equals to

(1) $\log_2 5$

(2) $1 - \log_2 5$

(3) $\log_5 2$

(4) None of the above

Question 30

The product of the distinct roots of $|x^2 - x - 6| = x + 2$ is

(1) –24

(2) –16

(3) –8

(4) –4

Question 31

Suppose, $\log_3 x = \log_{12} y = a$, where x, y are positive numbers. If G is the geometric mean of x and y, and $\log_6 G$ is equal to

(1) $\sqrt{a}$

(2) 2a

(3) a/2

(4) a

Question 32

If $\log(2^a \times 3^b \times 5^c)$ is the arithmetic mean of $\log(2^2 \times 3^3 \times 5)$, $\log(2^6 \times 3 \times 5^7)$, and $\log(2 \times 3^2 \times 5^4)$, then a equals

Question 33

If x is a positive quantity such that $2^x = 3^{\log_5 2}$, then x is equal to

(1) $\log_5 9$

(2) $\log_5 8$

(3) $1 + \log_3 \frac{5}{3}$

(4) $1 + \log_5 \frac{3}{5}$

Question 34

Let x and y be positive real numbers such that $\log_5 (x + y) + \log_5 (x - y) = 3$, and $\log_2 y - \log_2 x = 1 - \log_2 3$. Then xy equals

(1) 25

(2) 100

(3) 150

(4) 250

Question 35

If y is a negative number such that $2^{y^2 \log_3 5} = 5^{\log_2 3}$, then y equals

(1) $\log_2 (1/3)$

(2) $-\log_2 (1/5)$

(3) $-\log_2 (1/3)$

(4) $\log_2 (1/5)$

Thanks!